TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 1

5-MINUTE REVISION

Basics of Statistics

From the chapter's own revision points, formulae and questions.

BOOK-ONLY EDITION · Verified against the textbook

Selected from Chapter 1 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- A population is the complete group of interest; a sample is a part selected for study.
- A parameter describes a population, while a statistic is calculated from sample observations.
- The measurement scale determines which numerical comparisons are meaningful.
- Class intervals must cover the data without overlap; boundaries depend on the recording precision.
- Frequencies sum to the number of observations, and relative frequencies sum to one.

Quick revision · 2

- The source, selection method and reference period must be checked before interpreting data.

Important formulae · 1

Number of classes using Sturges' formula:

$$k = 1 + 3.322 \log_{10} N$$

Approximate class width:

$$C = \frac{|\max - \min|}{k}$$

A possible centring rule for a preliminary lower endpoint:

$$L = \min - \frac{C'k' - (\max - \min)}{2}$$

This is an optional starting value, not a necessary rule. Final limits and boundaries must respect the measurement precision, cover the data and avoid overlapping classes.

Important formulae · 2

Lower class boundary of the ith class:

$$L_i = \frac{\text{Lower class limit of } i + \text{Upper class limit of } (i-1)}{2}, \quad i = 2, 3, 4, \ldots$$

Upper class boundary of the ith class:

$$U_i = \frac{\text{Upper class limit of } i + \text{Lower class limit of } (i+1)}{2}, \quad i = 1, 2, 3, \ldots$$

Class mark:

$$\text{Class mark} = \frac{\text{Lower class limit} + \text{Upper class limit}}{2}$$

Important formulae · 3

Class width:

$$C = \text{Upper class boundary} - \text{Lower class boundary}$$

Relative frequency:

$$\text{Relative frequency} = \frac{f}{N}$$

Sum of frequencies:

$$\sum f = N$$

Important formulae · 4

Sum of relative frequencies:

$$\sum \frac{f}{N} = 1$$

Self-check from the chapter

1. The branch of statistics concerned with summarizing data is called _____ statistics. **[Descriptive]**
2. The branch of statistics concerned with drawing conclusions about a population from a sample is called _____ statistics. **[Inferential]**
3. A numerical constant describing a population is termed a _____. **[Parameter]**
4. A subset of a population selected for study is called a _____. **[Sample]**